Over-Blocking by the Legacy Content Script and the Redesigned Decision Logic

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Abstract, The content script shipped with SpoilFree 1.1 sent text from every page to the prediction API and blurred any text for which the server returned a positive flag. No watchlist, episode or strictness rules were applied, so unrelated text was blurred. This report documents the defect, the redesigned decision logic and the confidence bars that replaced the single server threshold.

Index Terms, content script, false positives, decision logic, spoiler detection, Chrome extension.

I. Introduction

The classifier only scores text. Whether a score leads to a blur is decided in the extension, which runs inside every page through a content script [1]. A model improvement therefore cannot remove false blurs if the decision rule is too permissive.

II. Observed Behaviour

The deployed content.js blurred any block of 40 to 500 characters for which the response field spoiler was true. The server sets that field at a threshold of 0.5 by default. The project file episode-parser.js was empty and the manifest loaded only content.js, so no episode logic could run. The store listing was at version 1.1, which allowed an old copy to run beside the development copy. Text that named no tracked title, including a chat message pasted from the developer's own conversation, was blurred.

III. Redesigned Decision Logic

A block is considered only if it names a tracked title or if the page title or address names one. The episode named in the text is parsed and compared with the stored progress, giving one of three states: seen, beyond or unknown. The bar applied to the returned confidence is shown below.

Episode already watched: never blurred.

Episode ahead of progress: confidence of at least 0.6.

Movie: confidence of at least 0.6.

No episode named, Strict on: at least 0.9 or 0.93 when only the page names the title.

No episode named, Strict off: never blurred.

Titles marked finished or inactive are ignored. Text containing "no spoilers" or "spoiler-free" is skipped. Progress, the finished flag and the strict flag are kept in local extension storage [2].

IV. Verification

The parser and matcher were exercised in Node with fixed inputs and the content script was run against a fake page with a stubbed API reply. After the update the developer reported that blurring behaved as intended. The training notebook proposes lower calibrated bars (see the training reports); those values were not adopted in the extension.

V. Conclusion

Moving the decision from a single server flag to a title-gated, episode-aware rule removed the unrelated blurs. The remaining wrong blurs on tracked titles depend on the model score.

References

[1] Google, "Content scripts," Chrome for Developers. Accessed: Oct. 28, 2026. [Online]. Available: https://developer.chrome.com/docs/extensions/develop/concepts/content-scripts

[2] Google, "chrome.storage," Chrome for Developers. Accessed: Oct. 28, 2026. [Online]. Available: https://developer.chrome.com/docs/extensions/reference/api/storage